Problem 7

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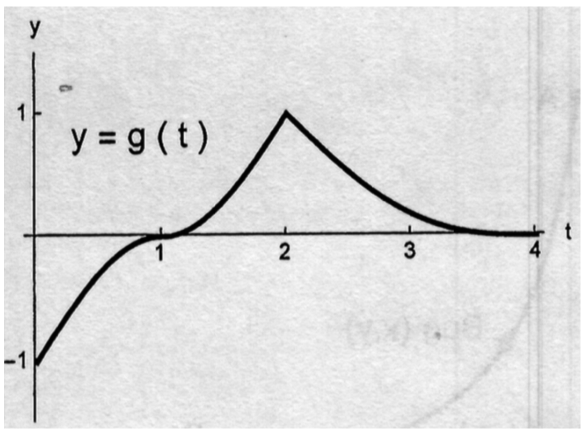
Problem 1 *The graph of g, a continuous function $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t)\,dt$ for $0 \le x \le 4$.*



(a) *Circle the correct statement about $A(1)$.*

- (i) $A(1) = 0$;
- (ii) $A(1) < 0$;
- (iii) $A(1) > 0$;
- (iv) *none of the previous answers.*

(b) *Circle the correct statement about $A(1.5)$.*

- (i) $A(1.5) = 0$;
- (ii) $A(1.5) < 0$;
- (iii) $A(1.5) > 0$;
- (iv) *none of the previous answers.*

(c) *Circle the correct statement about $A'(1.5)$.*

- (i) $A'(1.5) = 0$;
- (ii) $A'(1.5) < 0$;
- (iii) $A'(1.5) > 0$;

(iv) *none of the previous answers.*

(d) *Circle the correct expression for* $\int_1^4 (g(t)+1)\,dt$.

(i) $A(4)+1$;
(ii) $A(4)-A(1)$;
(iii) $A(4)-A(1)+1$;
(iv) $A(4)+3$;
(v) $A(4)-A(1)+3$;
(vi) *none of the previous answers.*

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(e) *Circle the correct statement about* $A(0)$.

(i) $A(0)=0$;
(ii) $A(0)=1$;
(iii) $0<A(0)<1$;
(iv) $A(0)>1$;
(v) $A(0)<0$;
(vi) *none of the previous answers.*

(f) *Circle the interval (or intervals) where the function* A *is DECREASING.*

(i) $(0,1)$;
(ii) $(1,2)$;
(iii) $(2,4)$;
(iv) *none of the previous answers.*

(g) *Circle the value (or values) where the function* A *attains its MAXIMUM.*

(i) $x=0$;
(ii) $x=1$;
(iii) $x=2$;
(iv) $x=3$;
(v) $x=4$;
(vi) *none of the previous answers.*

(h) *Circle the value (or values) where the function* A *attains its MINIMUM.*

(i) $x=0$;
(ii) $x=1$;

(iii) $x = 2$;
(iv) $x = 3$;
(v) $x = 4$;
(vi) *none of the previous answers.*

(i) *Circle the interval (or intervals) where the function A is CONCAVE DOWN.*

(i) $(0, 1)$;
(ii) $(1, 2)$;
(iii) $(2, 4)$;
(iv) *none of the previous answers.*

(j) *Circle the value (or values) of x where the function A has an inflection point.*

(i) $x = 1$;
(ii) $x = 2$;
(iii) $x = 3$;
(iv) *none of the previous answers.*

(k) *Sketch the graph of A, based on items (e)-(j)*